



Milestone 3

Individual Evidence Pack

- **College:** College of Computing
- **Major:** Computer Science
- **Group Number:** Project Team 2
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- **Milestone:** M3

Polymarket signal analysis — detect unusual odds movements empirically

1 What I Contributed

- I consolidated the sniper-detection output pipeline into a reproducible evidence package, including strict candidate ranking, top suspicious sessions, and verified case summaries for review.
- I curated and cross-checked the verified sniper cases (rank 2, 3, 4) by matching duration, buy/sell amounts, and anomaly scores between structured JSON evidence and generated attack-window plots.
- I consolidated the sniper-detection output pipeline into a reproducible evidence package, including strict candidate ranking, top suspicious sessions, and verified case summaries for review.
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2 Evidence

- Code interface evidence (sniper detection entry and case summary API: https://github.com/zrk494/ProjectOfPrivacy-enhancingTech/blob/main/submodules/algorithm/sniper_detection/src/detector.py)
- Configuration and reported module metrics in module documentation (54,582 orders, 27,188 sessions, 37 strict sniper candidates, 3 verified cases): https://github.com/zrk494/ProjectOfPrivacy-enhancingTech/blob/main/submodules/algorithm/sniper_detection/README.md
- Strict candidate output used in this milestone (41 lines including header, i.e., 40 candidate rows in current artifact version): https://github.com/zrk494/ProjectOfPrivacy-enhancingTech/blob/main/submodules/algorithm/sniper_detection/results/strict_sniper_candidates.csv
- Verified case detail records (rank 2/3/4, with tx hashes and buy/sell evidence): https://github.com/zrk494/ProjectOfPrivacy-enhancingTech/blob/main/submodules/visualize/frontend_sniper_detection/detailed_cases.json
- Plot evidence for verified sniper attack windows: https://github.com/zrk494/ProjectOfPrivacy-enhancingTech/blob/main/submodules/algorithm/sniper_detection/results/attack_windows/sniper_rank2_74seconds.png // https://github.com/zrk494/ProjectOfPrivacy-enhancingTech/blob/main/submodules/algorithm/sniper_detection/results/attack_windows/sniper_rank3_76seconds.png https://github.com/zrk494/ProjectOfPrivacy-enhancingTech/blob/main/submodules/algorithm/sniper_detection/results/attack_windows/sniper_rank4_72seconds.png

- Design artifact (architecture/interface contract used for static evidence integration):
<https://github.com/zrk494/ProjectOfPrivacy-enhancingTech/blob/main/submodules/visualize/docs/architecture.md>
- GitHub Repository: <https://github.com/zrk494/ProjectOfPrivacy-enhancingTech.git>

3 Validation Performed

What was tested:

What I validated: consistency between algorithm outputs and case-level evidence for verified sniper sessions.

Method:

I checked file-level result counts in the output artifacts:

- 「strict_sniper_candidates.csv」 has 41 lines including header, corresponding to 40 candidate rows in the current committed file.
- 「top10_suspicious_sessions.csv」 has 11 lines including header, corresponding to top 10 sessions.

I cross-validated rank 2/3/4 fields across 「detailed_cases.json」 and the detector summary logic (「get_case_summary」) for duration, buy/sell amounts, and anomaly scores.

I confirmed that each verified case has a matching attack-window figure in 「results/attack_windows/」 .

Result:

The evidence chain is internally consistent for all three verified cases